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Lab Practical #2:

1) Write a program to reverse a string using pointers.
2) Implement string functions like strlen, strcat, and strcmp.
3) Implement a program to read and write a text file.
4) Write a C program to test whether a number is prime or composite.
5) Write a C program to generate prime numbers in a given range.

Program-1:

```c
#include <stdio.h>

int main() {
    char str[100], *ptr_start, *ptr_end, temp;
    int len = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[len] != '\0' && str[len] != '\n')
        len++;

    ptr_start = str;
    ptr_end = str + len - 1;

    while (ptr_start < ptr_end) {
        temp = *ptr_start;
        *ptr_start = *ptr_end;
        *ptr_end = temp;

        ptr_start++;
        ptr_end--;
    }

    printf("Reversed string: %s\n", str);

    return 0;
}
```

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Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-2>gcc p1.c

D:\CSE B-Tech\Semester-6\INS\Lab-2>a
Enter a string: darshan
Reversed string: nahsrad
```

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Program-2:

```
#include <stdio.h>

int my_strlen(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        i++;
    }
    return i;
}

void my_strcat(char dest[], char src[]) {
    int i = 0, j = 0;

    while (dest[i] != '\0') {
        i++;
    }

    while (src[j] != '\0') {
        dest[i] = src[j];
        i++;
        j++;
    }

    dest[i] = '\0';
}

int my_strcmp(char s1[], char s2[]) {
    int i = 0;

    while (s1[i] != '\0' && s2[i] != '\0') {
        if (s1[i] != s2[i]) {
            return s1[i] - s2[i];
        }
        i++;
    }

    return s1[i] - s2[i];
}

int main() {
    char s1[100], s2[100];

    printf("Enter first string: ");
    fgets(s1, sizeof(s1), stdin);
```

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```
    printf("Enter second string: ");
    fgets(s2, sizeof(s2), stdin);

    for (int i = 0; s1[i]; i++)
        if (s1[i] == '\n') s1[i] = '\0';

    for (int i = 0; s2[i]; i++)
        if (s2[i] == '\n') s2[i] = '\0';

    printf("\nLength of s1 : %d\n", my_strlen(s1));
    printf("Length of s2 : %d\n", my_strlen(s2));

    my_strcat(s1, s2);
    printf("After concatenation (s1 + s2): %s\n", s1);

    int result = my_strcmp(s1, s2);

    if (result == 0)
        printf("Strings are equal.\n");
    else if (result > 0)
        printf("s1 is greater than s2.\n");
    else
        printf("s1 is smaller than s2.\n");

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-2>gcc p2.c

D:\CSE B-Tech\Semester-6\INS\Lab-2>a
Enter first string: darshan
Enter second string: marwadi

Length of s1 : 7
Length of s2 : 7
After concatenation (s1 + s2): darshanmarwadi
s1 is smaller than s2.
```

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Program-3:

```
#include <stdio.h>

int main() {
    FILE *fp;
    char text[200];

    fp = fopen("sample.txt", "w");

    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }

    printf("Enter text to write into file: ");
    fgets(text, sizeof(text), stdin);

    fprintf(fp, "%s", text);
    fclose(fp);

    fp = fopen("sample.txt", "r");

    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }

    printf("\nReading from file:\n");
    while (fgets(text, sizeof(text), fp) != NULL) {
        printf("%s", text);
    }

    fclose(fp);

    return 0;
}
```

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```
D:\CSE B-Tech\Semester-6\INS\Lab-2>gcc p3.c

D:\CSE B-Tech\Semester-6\INS\Lab-2>a
Enter text to write into file: hello

Reading from file:
hello
```

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Program-4:

```c
#include <stdio.h>

int main() {
    int num, i, flag = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num <= 1) {
        printf("Number is neither prime nor composite.\n");
        return 0;
    }

    // Check divisors from 2 to num/2
    for (i = 2; i <= num / 2; i++) {
        if (num % i == 0) {
            flag = 1;
            break;
        }
    }

    if (flag == 0)
        printf("%d is a PRIME number.\n", num);
    else
        printf("%d is a COMPOSITE number.\n", num);

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-2>gcc p4.c

D:\CSE B-Tech\Semester-6\INS\Lab-2>a
Enter a number: 24
24 is a COMPOSITE number.
```

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Program-5:

```
#include <stdio.h>

int main() {
    int start, end, i, j, flag;

    printf("Enter start of range: ");
    scanf("%d", &start);

    printf("Enter end of range: ");
    scanf("%d", &end);

    printf("Prime numbers between %d and %d are:\n", start, end);

    for (i = start; i <= end; i++) {
        if (i <= 1)
            continue;

        flag = 0;

        for (j = 2; j <= i / 2; j++) {
            if (i % j == 0) {
                flag = 1;
                break;
            }
        }

        if (flag == 0)
            printf("%d ", i);
    }

    printf("\n");
    return 0;
}
```

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```
D:\CSE B-Tech\Semester-6\INS\Lab-2>gcc p5.c

D:\CSE B-Tech\Semester-6\INS\Lab-2>a
Enter start of range: 34
Enter end of range: 87
Prime numbers between 34 and 87 are:
37 41 43 47 53 59 61 67 71 73 79 83
```